

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2020

Math 1113

(Differential and Integral Calculus)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any TWO questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define differentiability of a function. A function $f(x)$ is defined as follows: 18
- $$f(x) = 2x - 3 \quad \text{for } 0 \leq x \leq 2$$
- $$= x^2 - 3 \quad \text{for } 2 < x \leq 4$$
- Discuss the continuity and differentiability of $f(x)$ at $x = 2$.
- 1(b). If $u = \begin{vmatrix} x^2 & y^2 & z^2 \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix}$, then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. 12
- 2(a). Prove that the curves $x^{2/3} + y^{2/3} = c^{2/3}$ and $\frac{x^2}{2} + \frac{y^2}{2} = 1$ will touch each other 15
- if $a + b = c$.
- 2(b). Define asymptotes. 15
- Find the asymptotes of $x^3 + 3x^2y + 3xy^2 = x^2 + 2y^2 + 5x$
- 3(a). If $y = (x + \sqrt{x^2 + 1})^m$, then prove that 15
- $$(1+x^2) y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0.$$
- 3(b). Examine whether $x^{1/x}$ possesses a maximum or a minimum and determine it. 15

SECTION – B

4. Answer any two of the followings: 30
- a) $\int \frac{dx}{(1+x)\sqrt{(1+x-x^2)}}$
- b) $\int (x+3)\sqrt{(2x^2+2x+5)} dx$
- c) $\int \frac{dx}{25+\cos x+9\sin x}$
5. Answer any two of the followings: 30
- a). Evaluate $\int_4^{13} \frac{dx}{(x-5)\sqrt{(x+2)}}$
- b). $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{1}{8n} \right]$
- c). $\int_0^{\pi/2} \frac{x}{\sec x + \operatorname{cosec} x} dx$
- 6(a). Find the area above the x-axis included between the parabolas $y^2 = 5x$ and the circle $x^2 + y^2 = 10x$. 12
- 6(b). Evaluate $\int_0^\infty e^{-x^3} \cdot x^2 dx$ by using the rule of Gamma function. 08
- 6(c). Evaluate $\int_0^\pi \frac{dx}{(a+b \cos x)^2}$ by applying the rule of differentiation under the integral sign. 10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2020

Hum 1113

(Sociology and Behavioral Science)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any TWO questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is meant by sociology? What is the importance of sociology for the students of Energy Science and Engineering? 15
- 1(b). What is society and community? What are the basic differences between society and community? 15
- 2(a). What is meant by socialization? 05
- 2(b). Discuss the role of different organizations in socialization giving with examples. 25
- 3(a). What is crime and juvenile delinquency? 10
- 3(b). Discuss the causes of crime and juvenile delinquency in Bangladesh. 20

SECTION – B

- 4(a). What is organizational behavior? Describe the importance of studying organizational behavior for the students of Energy Science and Engineering. 15
- 4(b). What is training? Explain the importance of training needs assessment. 15
- 5(a). What is meant by employee relation? Explain the roles of an ideal employee. 15
- 5(b). What is leadership? Discuss the traits of effective leaders? 15
- 6(a). What is organizational design? Discuss the roles of organizational design. 15
- 6(b). Discuss the social impacts of unemployment. 15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2020

Ch 1113

(Chemistry-1)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any Two questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is SEI layer in lithium ion battery? Write down the reaction mechanism of lithium ion battery with appropriate diagram. 20
- 1(b). How does PEM fuel cell work? Explain. 10
- 2(a). Define nuclear binding energy. How does a nuclear reactor work? Explain with diagram. 20
- 2(b). If the average life of an atom is 5100 years, find the half life of that atom and also find the decay constant? 10
- 3(a). “Chemical equilibrium is a dynamic process” –Explain. 10
- 3(b). What are the law of mass action in chemical equilibrium? Explain. 10
- 3(c). “Galvanizing is better than tinning.” Explain. 10

SECTION – B

- 4(a). What is the degree of polymerization? Discuss the mechanism of free radical addition polymerization. 13
- 4(b). How would you differentiate thermosetting polymer from thermoplastic polymer? 09
- 4(c). A polymer mixture contains two polymers, one having molecular weight 10,0000 and the other having molecular weight 80,000. The two components are present in equimolar concentration. Calculate the number averages and weight average molecular weight. 08
- 5(a). What is heavy metal? Discuss the biochemical effects of arsenic. 12
- 5(b). How does hydrocarbon play a role in formation of photochemical Smog? 10
- 5(c). Why COD is higher than BOD? 08
- 6(a). What are the differences among coke, coal and charcoal? 09
- 6(b). What is octane number of a fuel? How will you improve the octane number of fuel? 14
- 6(c). Write down the characteristic properties of a good fuel. 07

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2020

Ph 1113

(Physics)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any TWO questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define progressive wave and stationary wave. Distinguish between them. 06
- 1(b). Discuss mathematically the phenomenon of forced vibration. 16
- 1(c). Show that for a particle executing simple harmonic motion, the instantaneous velocity is $\omega\sqrt{a^2 - y^2}$ and instantaneous acceleration is $-\omega^2 y$. 08
- 2(a). Draw the following planes and directions in a cubic crystal: 10
(1 1 0), (0 2 0), $(\bar{1} \bar{1} 1)$, [0 2 1], [0 2 0].
- 2(b). Show that at large temperature Einstein model gives same result as classical theory does. 15
- 2(c). Show that the packing fraction for *fcc* lattice is 0.74. 05
- 3(a). Why do we say that Schrödinger equation is an eigen value equation? 08
Using Hamiltonian operator derive an expression for Schrödinger equation in one dimension.
- 3(b). What is Hall effect? Deduce an expression for Hall coefficient for a metal 14
and show the relation between Hall coefficient and Hall mobility.
- 3(c). Copper has a mass density of 8.9 gm/cm³ and electric resistivity of 1.5 × 10⁻⁸ ohm-m at room temperature. Calculate Fermi energy and Fermi velocity. 08

SECTION – B

- 4(a). An electron is not the part of a nucleus. Is it true? Give logic in favor of your statement. 08
- 4(b). Show that $\lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos\phi)$ 16
Where the symbols have their usual meanings.
- 4(c) Find a relation between phase velocity and group velocity. 06
- 5(a). Define binding energy and packing fraction. Draw the graphs for them and explain the graphs. 12
- 5(b). Write down three sets of quantum numbers of electron for $n=3$. 10
- 5(c) How much energy is required to remove an electron in the $n=2$ state from a hydrogen atom? 08
- 6(a). What is meant by biological effects of radiation? Distinguish between (i) chronic dose and acute dose, (ii) somatic effect and genetic effect, and (iii) acute effect and latent effect. 16
- 6(b). Explain the properties of α , β and γ . 06
- 6(c) Calculate how much energy liberate from 1 kg of ${}^{235}_{92}\text{U}$. 08

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2020

ESE 1101

(Fundamentals of Energy Resources)

Time: 1 Hour 30 Minutes.

Full Marks: 120

- N.B. i) Answer any TWO questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

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| 1(a). | What is meant by energy resources? How do energy resources are classified globally? Explain in brief. | 10 |
| 1(b). | Differentiate between non-renewable and renewable energy sources. What are the adverse effects of using non-renewable energy sources? | 10 |
| 1(c). | Distinguish between primary energy and secondary energy resources. Write a note on non-conventional energy resources. | 10 |
| 2(a). | Write a short note on history of energy uses. | 08 |
| 2(b). | Briefly discuss about the current status of conventional energy sources of Bangladesh. | 10 |
| 2(c). | What are the different grades of coal? Write and explain the steps of coal exploration process. | 12 |
| 3(a). | Classify the petroleum oil. Briefly discuss about the Biogenic theory of formation of petroleum oil. | 10 |
| 3(b). | What is natural gas and how they are formed? Write in brief the potential of NG in Bangladesh. | 10 |
| 3(c). | Define oil shale and tar sand. Write down the advantages and disadvantages of Nuclear energy. | 10 |

SECTION – B

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|-------|--|----|
| 4(a). | What is solar energy and how this energy is formed? Why do radiation of earth surface is less than that at the extraterrestrial region? Explain. | 10 |
| 4(b). | Briefly explain the thermal application of solar energy. With schematic diagram explain the working principle of a solar water heating system with a circulation pump. | 10 |
| 4(c). | Differentiate between thermal conversion and photovoltaic conversion of solar energy. | 06 |
| 4(d) | What are the disadvantages of solar energy? | 04 |
| | | |
| 5(a). | Distinguish between biomass and bioenergy. Write five advantages and five disadvantages of using biomass. | 10 |
| 5(b). | Why do conversion of biomass is necessary? Explain any method of obtaining liquid fuel from biomass. | 12 |
| 5(c) | Define the gasification process. Why biomass conversion to bioenergy is beneficial for environment? Explain. | 08 |
| | | |
| 6(a). | Describe the classification of hydroelectric power plants. How do pump storage plants works? | 10 |
| 6(b). | What factors are considered for site selection of a wind energy conversion system? Explain with necessary diagrams. | 10 |
| 6(c). | How to extract the energy from geothermal energy resources? Write down the disadvantages of geothermal energy resources. | 10 |

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